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Q. What is DTMF and how does it work? What are the applications of DTMF?

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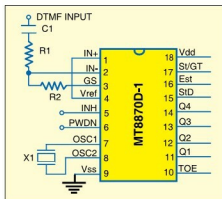
A. Dual-tone multi-frequency (DTMF) is used for signaling over analogue telephone lines in the voice-frequency band. DTMF signaling is used between telephone handsets and other communications devices and the switching centre. DTMF used in pushbutton telephones for tone dialing is known as touch-tone DTMF. It

TABLE I
Tone and Key Assignments

Frequencies	1209 Hz	1336 Hz	1477 Hz	1633 Hz
697 Hz	1	2	3	A
770 Hz	4	5	6	B
852 Hz	7	8	9	C
941 Hz	*	0	#	D

TABLE II
Special Tone Frequencies

Event	Low frequency	High frequency
Busy signal	480 Hz	620 Hz
Ring back tone	440 Hz	480 Hz
Dial tone	350 Hz	440 Hz



Pin assignment of MT8870 DTMF decoder

uses a mix of two pure tones (pure sine waves). The system uses eight different frequency signals transmitted in pairs to represent 16 different numbers, symbols and letters.

The DTMF keypad is laid out in a 4x4 matrix, with each row representing a low frequency and each column

representing a high frequency. DTMF assigns a specific frequency (consisting of two separate tones) to each key so that it can easily be identified by a microprocessor or other devices. The dual tones are the reason for calling the system dual-tone multi-frequency. These tones are then decoded by the switching centre to determine which key was pressed. The signal generated by a DTMF encoder is a direct algebraic summation, in real time, of the amplitudes of two sine (cosine) waves of different frequencies, i.e., pressing '5' will send a tone made by adding 1336 Hz and 770 Hz to the other end of the line. Table I shows the tones and key assignments in a DTMF system.

Special tone frequencies to indicate the status of lines, equipment or the result of calls are shown in Table II.

The tone frequencies, as defined by the Precise Tone Plan, are selected such that harmonics and inter modulation products do not cause an unreliable signal. No frequency is a multiple of another, the difference between any two frequencies does not equal any of the frequencies, and the sum of any two frequencies does not equal any of the frequencies.

Pin assignment of the MT8870 series DTMF-to-BCD decoder chip is shown in the figure. Input signals are given at pin 2 (IN-) of the chip. The correct 4-bit decoded signal of the DTMF tone is transferred to output pins 11 through 14 (Q1 through Q4).

DTMF signaling is also used in remote monitoring and control of applications. DTMF outputs can be used with relay drivers for pan/tilt/focus/zoom control of video cameras.

Answers compiled by Col (Retd) N.C. Pande

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